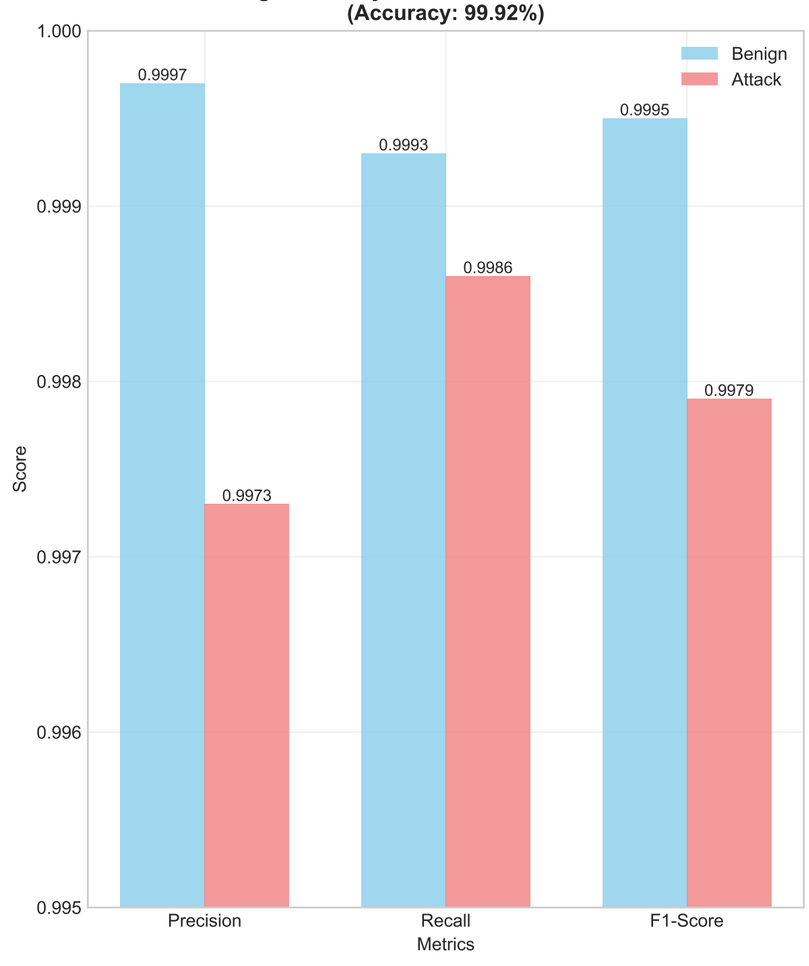
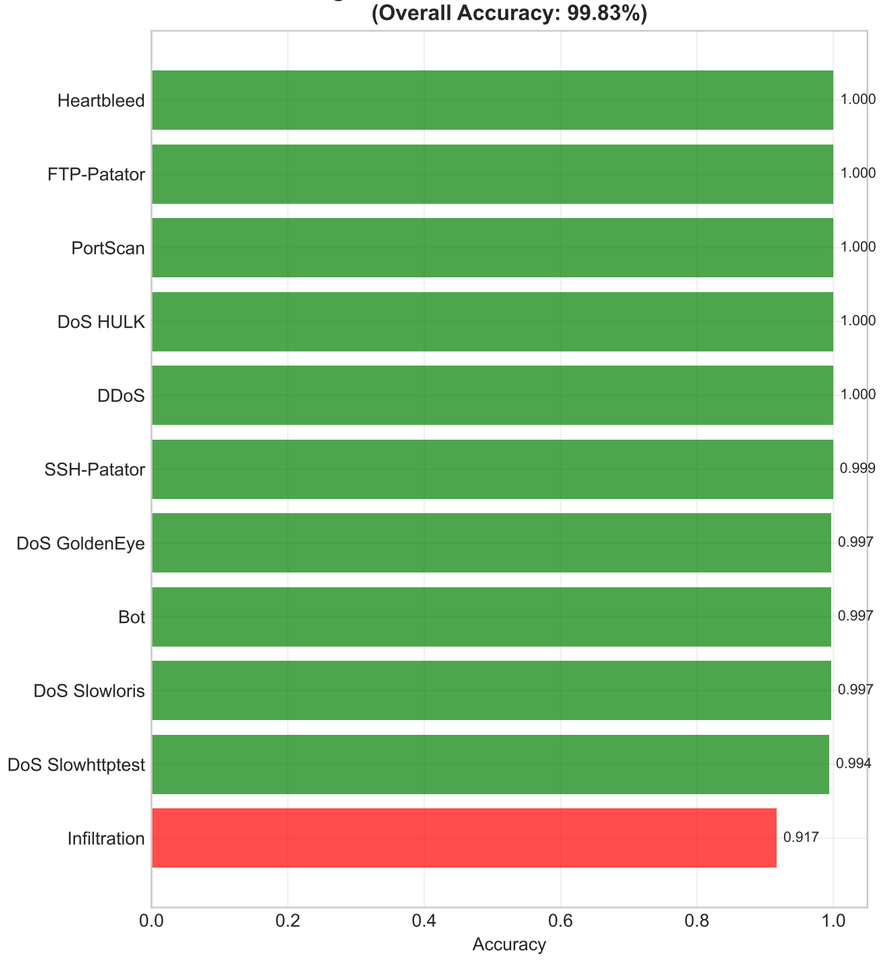
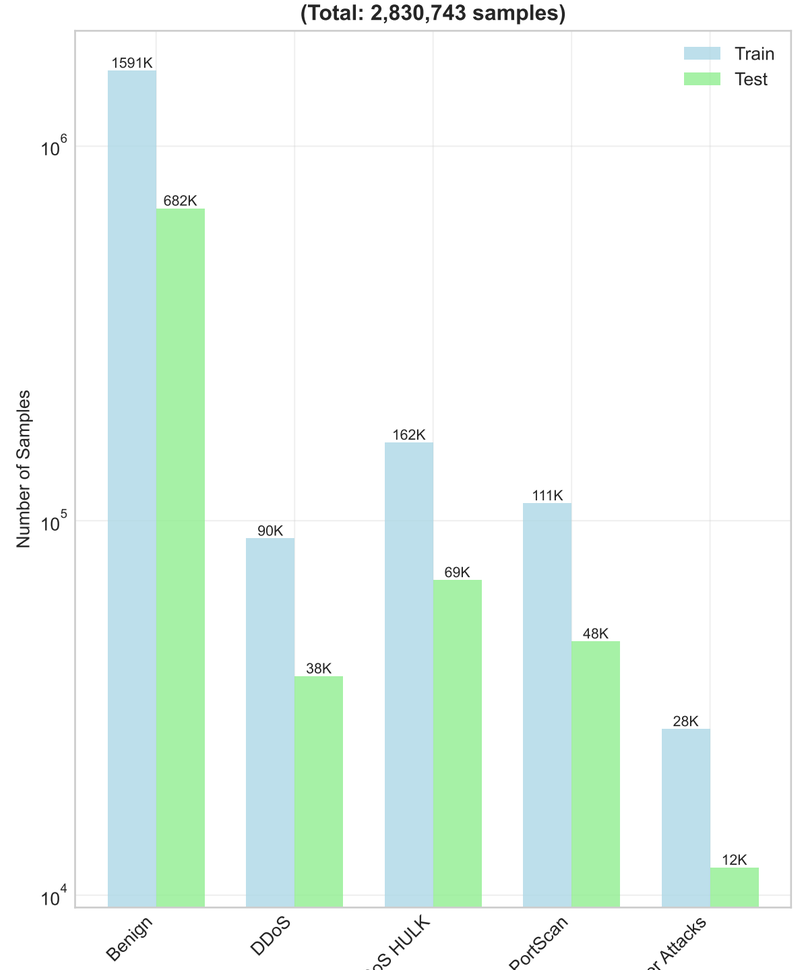
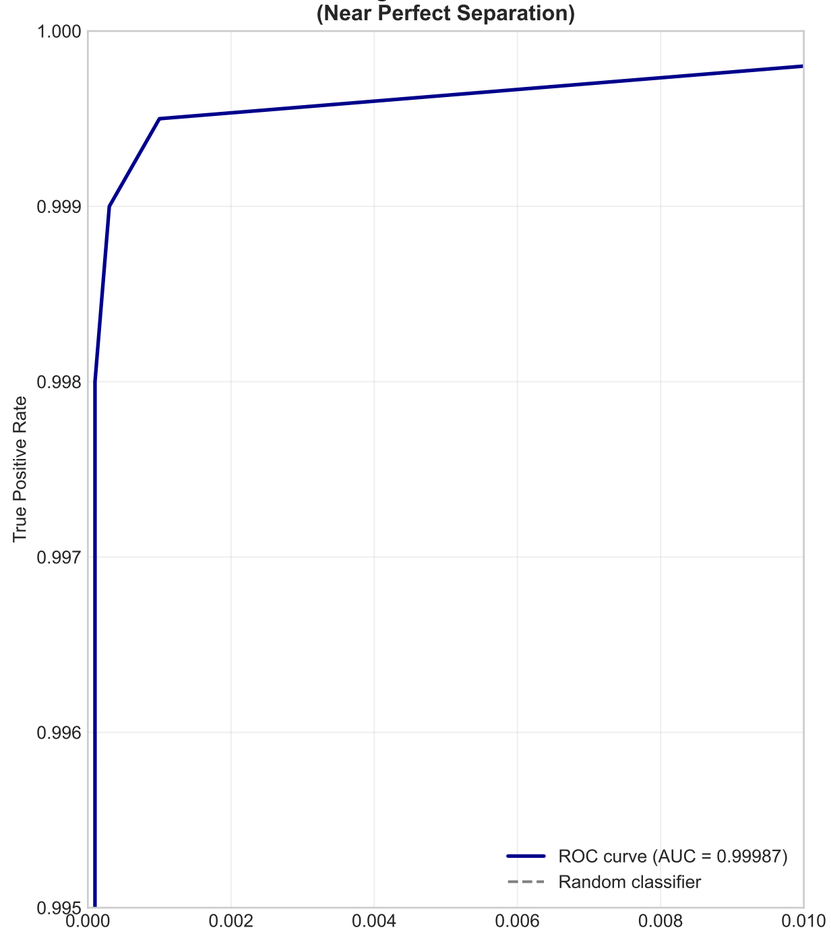
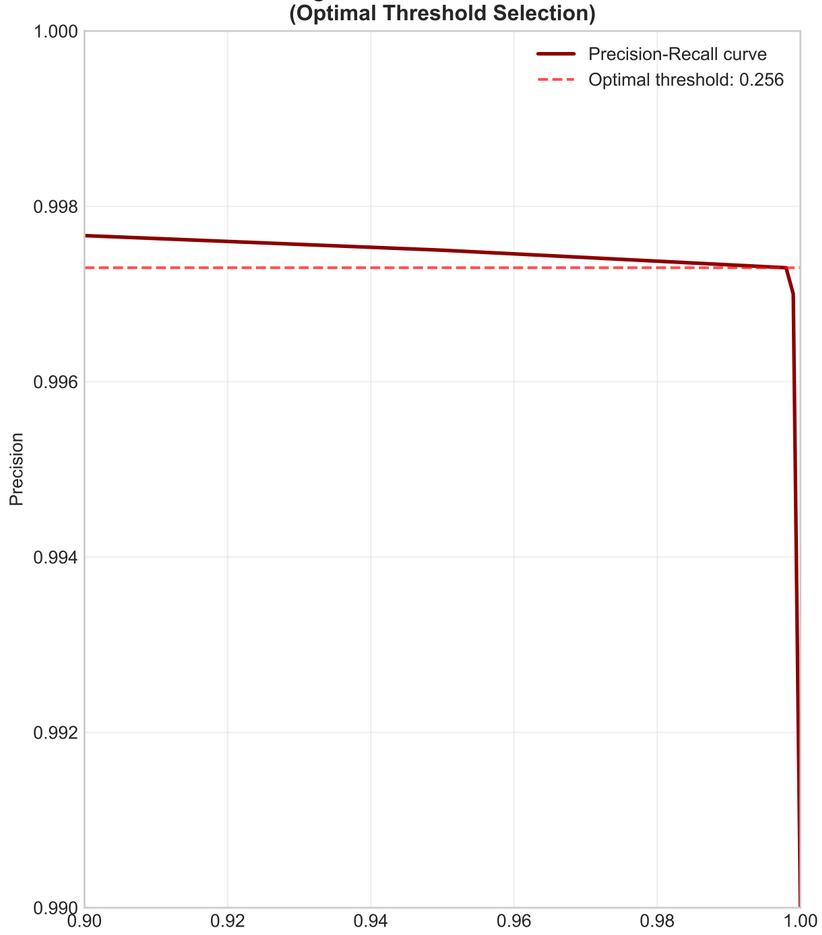
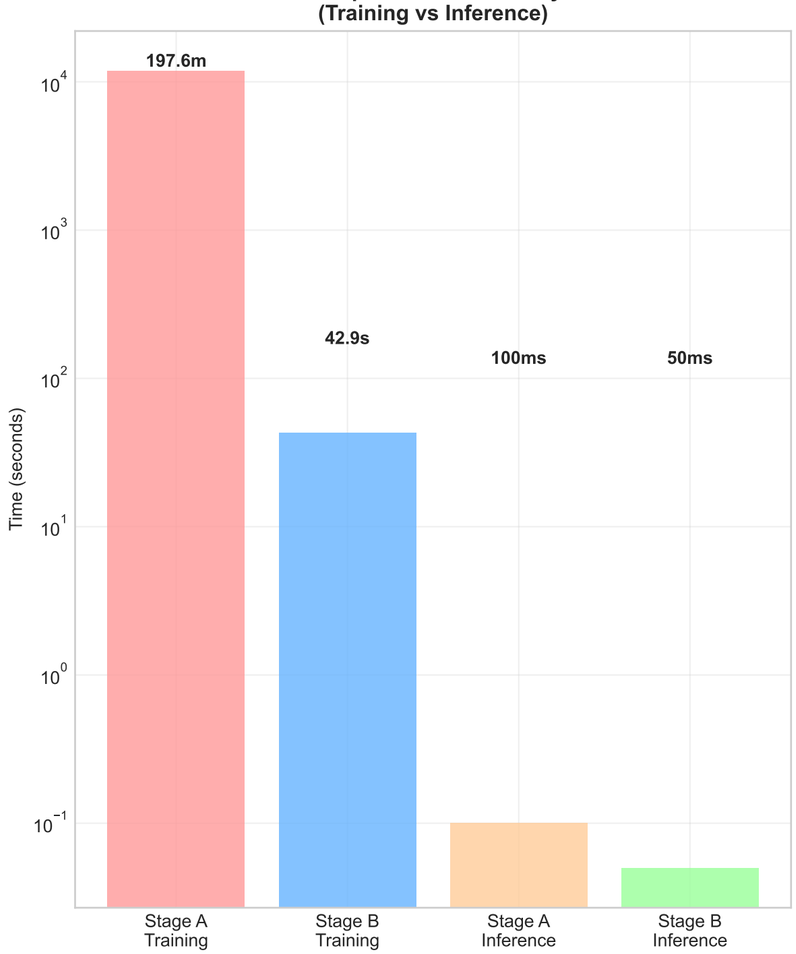

Stage A: Binary Classification Performance
(Accuracy: 99.92%)
Benign
Attack
0.9997
0.9973
0.9993
0.9986
0.9995
0.9979
Score
Precision
Recall
F1-Score
Metrics


Stage B: Multi-class Attack Classification
(Overall Accuracy: 99.83%)
Heartbleed
FTP-Patator
PortScan
DoS HULK
DDoS
SSH-Patator
DoS GoldenEye
Bot
DoS Slowloris
DoS Slowhttptest
Infiltration
1.000
1.000
1.000
1.000
1.000
0.999
0.997
0.997
0.997
0.994
0.917
Accuracy


Dataset Distribution
(Total: 2,830,743 samples)
Train
Test
1591K
682K
90K
38K
162K
69K
111K
48K
28K
12K
Number of Samples
Benign
DDoS
DoS HULK
PortScan
Other Attacks
Traffic Type


Stage A: ROC Curve
(Near Perfect Separation)
ROC curve (AUC = 0.99987)
Random classifier
True Positive Rate
False Positive Rate


Stage A: Precision-Recall Curve
(Optimal Threshold Selection)
Precision-Recall curve
Optimal threshold: 0.256
Precision
Recall


Computational Efficiency
(Training vs Inference)
197.6m
42.9s
100ms
50ms
Time (seconds)
Stage A
Training
Stage B
Training
Stage A
Inference
Stage B
Inference